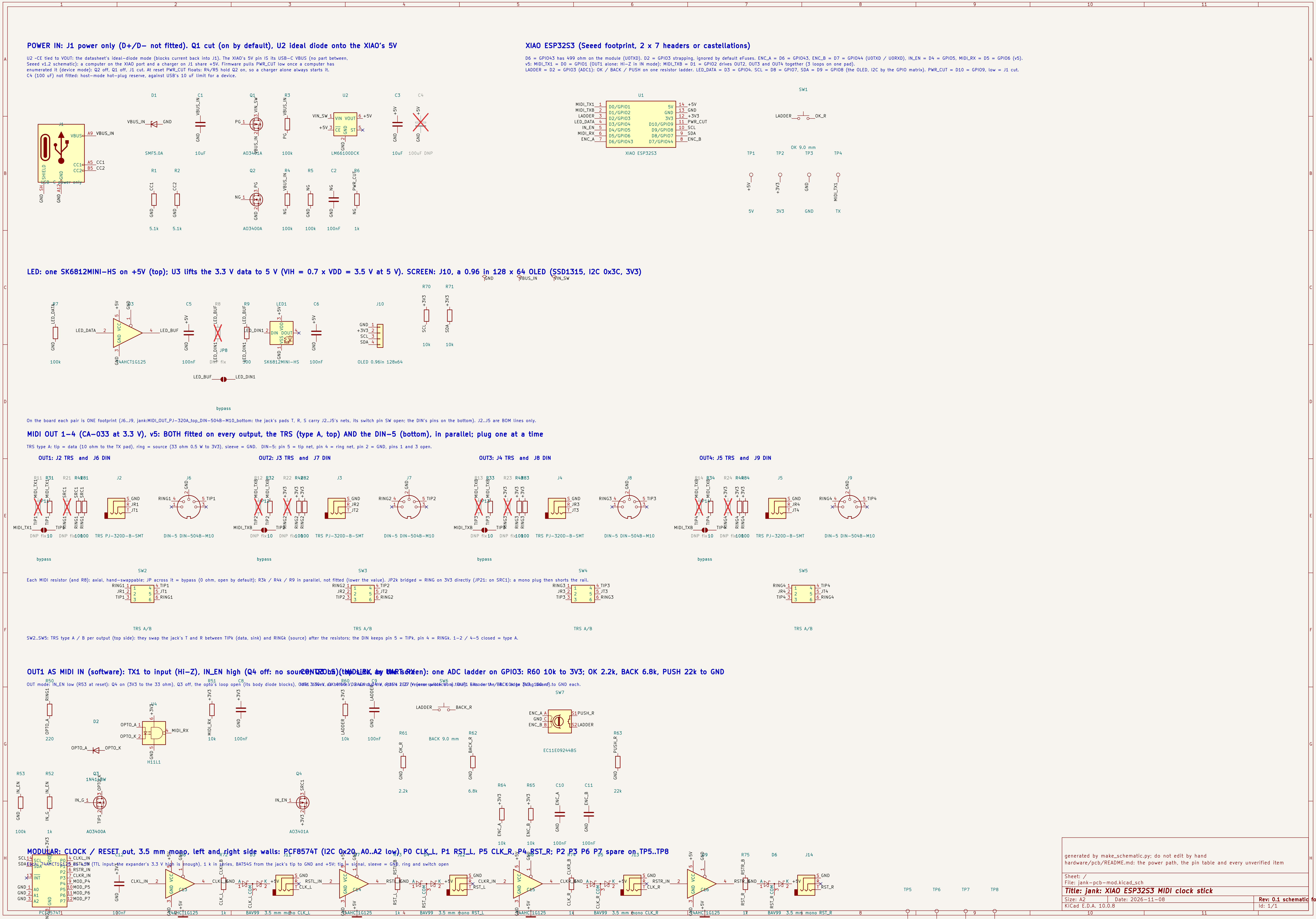
J1
VBUS_IN
CC1
CC2
USB-C power only
D1
SMF5.0A
C1
10uF
Q1
AO3401A
R3
100k
U2
LM66100DCK
C3
10uF
C4
100uF DNP
R1
5.1k
R2
5.1k
Q2
AO3400A
R4
100k
R5
100k
C2
100nF
R6
1k
U1
XIAO ESP32S3
SW1
OK 9.0 mm
TP1 5V
TP2 3V3
TP3 GND
TP4 TX
R7
100k
U3
74AHCT1G125
C5
100nF
R8
DNP fix
JP8
bypass
R9
300
LED1
SK6812MINI-HS
C6
100nF
J10
OLED 0.96in 128x64
R70
10k
R71
10k
J2 TRS PJ-320D-B-SMT
J6 DIN-5 DIN-504B-M10
J3 TRS PJ-320D-B-SMT
J7 DIN-5 DIN-504B-M10
J4 TRS PJ-320D-B-SMT
J8 DIN-5 DIN-504B-M10
J5 TRS PJ-320D-B-SMT
J9 DIN-5 DIN-504B-M10
SW2 TRS A/B
SW3 TRS A/B
SW4 TRS A/B
SW5 TRS A/B
R50
220
D2
U4
H11L1
R51
10k
C8
100nF
R60
10k
C9
100nF
SW6
BACK 9.0 mm
SW7
EC11E09244BS
R61
2.2k
R62
6.8k
R63
22k
R53
100k
R52
1k
Q3
1N4148W
AO3400A
Q4
AO3401A
R64
10k
R65
10k
C10
100nF
C11
100nF
U5
PCF8574T
C12
100nF
U6 74AHCT1G125
R72 1k
D3 BAV99
J11 3.5 mm mono CLK_L
U7 74AHCT1G125
R73 1k
D4 BAV99
J12 3.5 mm mono RST_L
U8 74AHCT1G125
R74 1k
D5 BAV99
J13 3.5 mm mono CLK_R
U9 74AHCT1G125
R75 1k
D6 BAV99
J14 3.5 mm mono RST_R
TP5
TP6
TP7
TP8

POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal−diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB−C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it.
C4 (100 uF) not fitted: host−mode hot−plug reserve, against USB's 10 uF limit for a device.

XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D4 = GPIO5, MIDI_RX = D5 = GPIO6 (v5).
v5: MIDI_TX1 = D0 = GPIO1 (OUT1 alone: Hi−Z in IN mode); MIDI_TXB = D1 = GPIO2 drives OUT2, OUT3 and OUT4 together (3 loops on one pad).
LADDER = D2 = GPIO3 (ADC1): OK / BACK / PUSH on one resistor ladder. LED_DATA = D3 = GPIO4. SCL = D8 = GPIO7, SDA = D9 = GPIO8 (the OLED, I2C by the GPIO matrix). PWR_CUT = D10 = GPIO9, low = J1 cut.

LED: one SK6812MINI−HS on +5V (top); U3 lifts the 3.3 V data to 5 V (VIH = 0.7 x VDD = 3.5 V at 5 V). SCREEN: J10, a 0.96 in 128 x 64 OLED (SSD1315, I2C 0x3C, 3V3)

On the board each pair is ONE footprint (J6..J9, jank:MIDI_OUT_PJ−320A_top_DIN−504B−M10_bottom: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; the DIN's pins on the bottom). J2..J5 are BOM lines only.

MIDI OUT 1−4 (CA−033 at 3.3 V), v5: BOTH fitted on every output, the TRS (type A, top) AND the DIN−5 (bottom), in parallel; plug one at a time

TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN−5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.

OUT1: J2 TRS and J6 DIN

OUT2: J3 TRS and J7 DIN

OUT3: J4 TRS and J8 DIN

OUT4: J5 TRS and J9 DIN

Each MIDI resistor (and R8): axial, hand−swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.

SW2..SW5: TRS type A / B per output (top side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1−2 / 4−5 closed = type A.

OUT1 AS MIDI IN (software): TX1 to input (Hi−Z), IN_EN high (Q4 off: no sour[illegible]

OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). [illegible]

CONTROLS [illegible]): one ADC ladder on GPIO3: R60 10k to 3V3; OK 2.2k, BACK 6.8k, PUSH 22k to GND

[illegible] 10k to 3V3, 100 nF to GND each.

MODULAR: CLOCK / RESET out, 3.5 mm mono, left and right side walls: PCF8574T (I2C 0x20, A0..A2 low) P0 CLK_L, P1 RST_L, P5 CLK_R, P4 RST_R; P2 P3 P6 P7 spare on TP5..TP8

Each: 74AHCT1G125 on +5V (TTL input: the expander's 3.3 V high is enough), 1 k in series, BAT54S from the jack's tip to GND and +5V; tip = signal, sleeve = GND, ring and switch open

generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank−pcb−mod.kicad_sch

Title: jank: XIAO ESP32S3 MIDI clock stick

Size: A2
Date: 2026−11−08
Rev: 0.1 schematic for review
KiCad E.D.A. 10.0.8
Id: 1/1